

Q1. Giving reasons for your answer based on the type of system being developed, **suggest** [06] CLO4 the most appropriate generic software process model that might be used as a basis for managing the development of the following systems:

- a. A system to control anti-lock braking in a car — *spiral*
- b. An interactive travel planning system that helps users plan journeys with the lowest environmental impact. — *prototyping*

Q2. Outline Software Requirement Engineering Process, with short description of each stage [05] CLO2 also **Determine** why it is important to make a distinction between developing the user requirements and developing system requirements in the requirements engineering process.

Q3. List any 8 principles of agile methodologies and **Explain** how these principles leads to [05] CLO1 the accelerated development and deployment of software.

Q4. You are software engineer at a company where management routinely encourages you [04] CLO3 and your colleagues to use pirated software. How would you respond in a way that is legal, moral and ethical?

- Q01. Propose a specific software project that would be amenable to prototyping model [04] CLO3
present scenario for applying model to software.
- Q02. List any 8 Agile software development principles [04] CLO1
- Q03. Outline Software Requirement Engineering Process; briefly describe each stage, its relation to other stage and its overall importance. [05] CLO2
- Q04. You are the owner of the software engineering company. Your employees (engineers) want you to pay for them to attend training. How would you respond in a way that is legal, moral and ethical? [02] CLO3
- Q05. Describe scrum in detail [05] CLO1

5) Highly attention to technological excellence and good designs

6) Working teams are motivated individual.

7) Simplicity maximizing amount of work not done is essential.

8) Regularity team reflects which increases the efforts and adjust accordingly.

Q: 3 CLO2

Software Requirement System

Limitations, functions, behaviour and features expectations gathered from

Review (PTR). They check and make sure ^{if} that all the requirements are fulfilled, ~~are~~ are realistic, are consistent, could be checked for validation.

After validation, no change is to be made in the requirements (such as in waterfall, V and V) but if a change is to be made. The entire process is followed in the same way.

7. Requirement Management

This management is carried out in whole process of SRE. It is supposed to manage any changes made and the traceability.

Importance of SRE Process

Software Requirement engineering process allows you developer to gather requirement in a systematic way, making it difficult to miss ~~an~~ requirements. This results in strong foundation of software and ease in future development of software for both developer and customer.

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Q#4

I am the owner of a software company and I would want my employees to be satisfied with my company so they work with the full capacity and skills.

However, as I run a company I could not provide extra facilities. If my employees want me to pay for their training I would simply show them my company's policies that do not involve that owner is supposed to do what they want me to. I would let them know that employee's training is not company's responsibility. and simply professionally refuse/decline their call.

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SCRUM

An agile methodology exclusively helpful for a project management. Scrum is suitable if the project has aggressive deadlines, complexity and higher degree of uniqueness. Scrum allows a team to work in a proper way.

The three main roles in Scrum are

Scrum Master

leader and manager of the team. Team's performance is his responsibility.

Team (5-9)

5 to 9 members form up a team and they are supposed to fulfill all the needs of a project. ~~Team~~ development team such as ex: requirements engineers, developers, designers etc.

Product Owner.

A key stakeholder (suitable for each process), may change if requirements are from different sections.

⇒ **Product Backlog**

Contains all the requirements. These requirements may be divided/categorized and then filtered.

⇒ Sprint Backlog

Contains the categorized requirement for one iteration (sprint). These may also be refined.

+ Sprint Planning

Requirements for ~~sp~~ product backlog are planned, filtered, ~~refined and~~ then the working is planned of Sprint backlog (requirement for one single iteration). This divides all the jobs to be done.

+ ~~Sprint~~ Daily Scrum

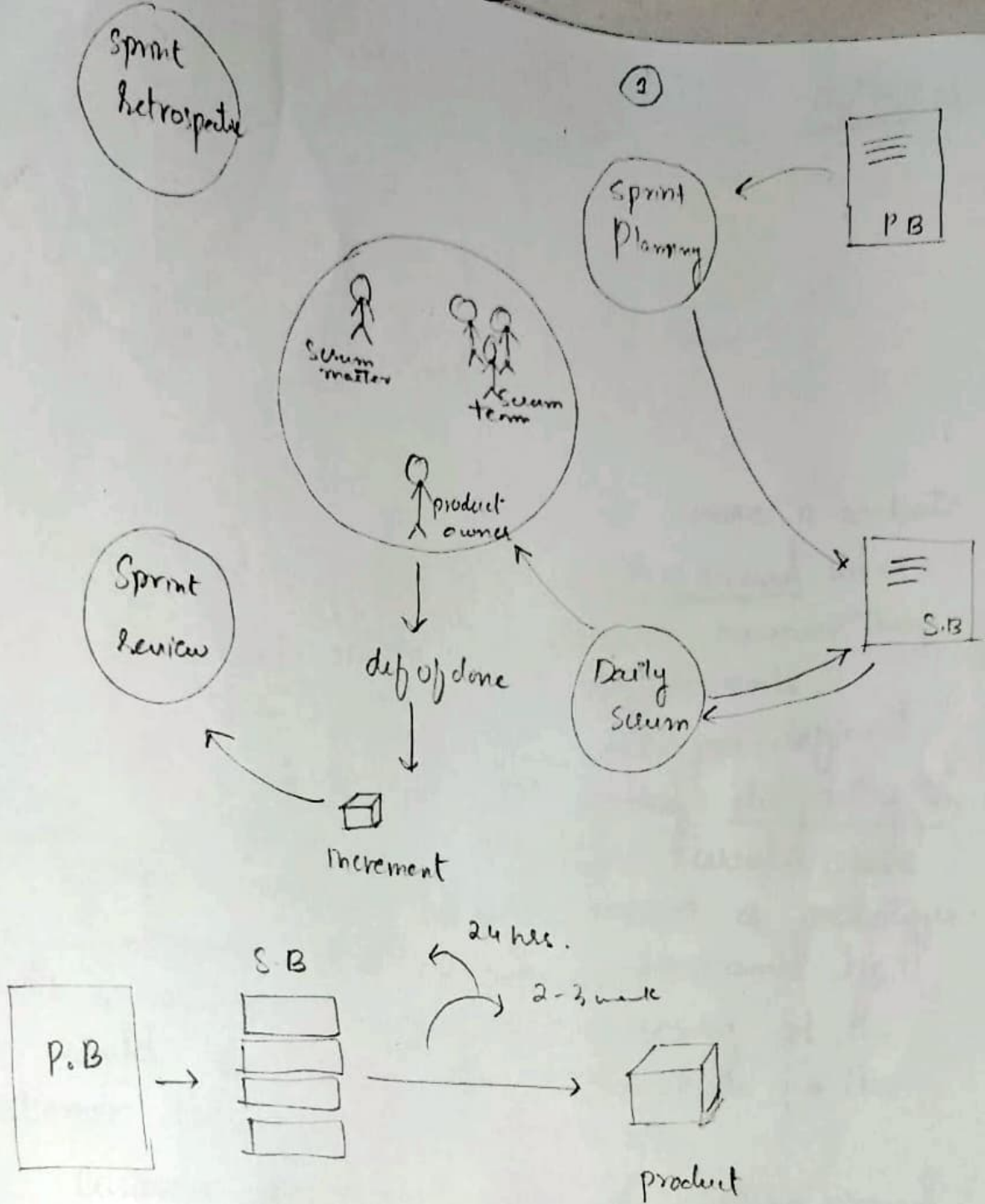
The entire team holds a meeting and studies what needs to be done, brainstorm together, ~~may also do work on backlog~~. They discuss all the things done in one day daily.

+ Sprint Review

After a sprint is completed and there is an increment, entire team sit together, holds a meeting and reviews the sprint together. This may not involve ~~so~~ customer's presence.

+ Sprint Retrospective

A meeting ~~is~~ arranged to review the sprints with customer. This allows customer feedback. Customer may propose changes that are than accommodated in next sprint.



Problems with Scrum

- + Scrum team plays a major role (as it is an agile methodology). If the team is involved in conflicts, it will affect the whole process.
- + Repeated discussions might cause in lack of interest and may feel boring.